3.4.5 Species and taxonomy

Content	Opportunities for skills development
Two organisms belong to the same species if they are able to produce fertile offspring. Courtship behaviour as a necessary precursor to successful mating. The role of courtship in species recognition. A phylogenetic classification system attempts to arrange species into groups based on their evolutionary origins and relationships. It uses a hierarchy in which smaller groups are placed within larger groups, with no overlap between groups. Each group is called a taxon (plural taxa). One hierarchy comprises the taxa: domain, kingdom, phylum, class, order, family, genus and species. Each species is universally identified by a binomial consisting of the name of its genus and species, eg, *Homo sapiens*. Recall of different taxonomic systems, such as the three domain or five kingdom systems, will **not** be required. **Students should be able to** appreciate that advances in immunology and genome sequencing help to clarify evolutionary relationships between organisms.	